DTC P0089:00 [SKYACTIV-G 1.3, SKYACTIV-G 1.5]

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Note
- To determine the malfunctioning part, proceed with the diagnostics from "Function Inspection Using M-MDS".

Details On DTCs

DESCRIPTION	Spill valve control solenoid valve control circuit range/performance problem	
DETECTION CONDITION	Determination conditions	• The average fuel pressure on the high pressure side, measured by the fuel pressure sensor, exceeds the specified value.
	Preconditions	• Engine speed: **500 rpm or more** *1 • The following DTCs are not detected: — Fuel pressure sensor: P0087:00, P0088:00, P0192:00, P0193:00 — High pressure fuel pump: P0091:00, P0092:00 *1: Value can be verified by displaying PIDs using M-MDS
	Malfunction determination period	• **2 s** period
	Drive cycle	• 1
	Self test type	• CMDTC self test, KOER self test
	Sensor used	• Fuel pressure sensor
FAIL-SAFE FUNCTION	• Limits intake air amount • Stops high pressure fuel pump control. • The upper limit of the engine speed is **2,500 rpm**.	
VEHICLE STATUS WHEN DTCs ARE OUTPUT	• Illuminates check engine light.	
POSSIBLE CAUSE	• High pressure fuel pump connector or terminals malfunction • PCM connector or terminals malfunction • Fuel pressure sensor malfunction • High pressure fuel pump malfunction — Spill valve control solenoid valve (built-into high pressure fuel pump) malfunction • High pressure fuel pipe clogging • Fuel distributor malfunction • Fuel injector malfunction • PCM malfunction	

System Wiring Diagram
- Not applicable

Function Explanation (DTC Detection Outline)
- The PCM calculates the target fuel pressure appropriate to the engine conditions relative to the actual fuel pressure based on the fuel pressure signal, and provides feedback to the high pressure fuel pump control. If the average fuel pressure exceeds the specified value, the PCM determines a malfunction in the high pressure fuel pump system, and stores a DTC.

Repeatability Verification Procedure
1. Warm up the engine to allow the engine coolant temperature to reach **80 °C {176 °F} or more**.
2. Start the engine and leave it idling for **1 min**.

PID Item/Simulation Item Used In Diagnosis
- Not applicable

Function Inspection Using M-MDS

STEP	INSPECTION	RESULTS	ACTION
1	**PURPOSE: VERIFY RELATED SERVICE INFORMATION AVAILABILITY** • Verify related Service Information availability. • Is any related Service Information available?	Yes	Perform repair or diagnosis according to the available Service Information. • If the vehicle is not repaired, go to the next step.
		No	Go to the next step.

STEP	INSPECTION	RESULTS	ACTION
2	**PURPOSE: IDENTIFY TRIGGER DTC FOR FREEZE FRAME DATA** • Is the DTC P0089:00 on FREEZE FRAME DATA?	Yes	Go to the next step.
		No	Go to the troubleshooting procedure for DTC on FREEZE FRAME DATA. (See DTC TABLE [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
3	**PURPOSE: RECORD VEHICLE STATUS AT TIME OF DTC DETECTION TO UTILIZE WITH REPEATABILITY VERIFICATION** **Note** • Recording can be facilitated using the screen capture function of the PC. • Record the FREEZE FRAME DATA/snapshot data on the repair order.	—	Go to the next step.
4	**PURPOSE: VERIFY RELATED PENDING CODE AND/OR DTC** • Switch the ignition off, then ON (engine off). • Perform the Pending Trouble Code Access Procedure and DTC Reading Procedure. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) • Are any other PENDING CODEs and/or DTCs present?	Yes	Go to the applicable PENDING CODE or DTC inspection. (See DTC TABLE [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) Go to the troubleshooting procedure to perform the procedure from Step 1.
		No	Go to the troubleshooting procedure to perform the procedure from Step 1.

Troubleshooting Diagnostic Procedure
Intention of troubleshooting procedure

- Step 1
  - — Replace the high pressure fuel system parts.
- Step 2—3
  - — Verify that the primary malfunction is resolved and there are no other malfunctions.

STEP	INSPECTION	RESULTS	ACTION
1	**PURPOSE: REPLACE HIGH PRESSURE FUEL SYSTEM RELATED PARTS DUE TO MALFUNCTION IN HIGH PRESSURE FUEL SYSTEM PARTS** • Replace the following parts. — High pressure fuel pump (See HIGH PRESSURE FUEL PUMP REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) — High pressure fuel pipe (See HIGH PRESSURE FUEL PUMP REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) — Fuel injector No.1 (See FUEL INJECTOR REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) — Fuel injector No.2 — Fuel injector No.3 — Fuel injector No.4 — Fuel distributor (See FUEL INJECTOR REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)	—	Go to the next step.

STEP	INSPECTION	RESULTS	ACTION
2	**PURPOSE: VERIFICATION OF VEHICLE REPAIR COMPLETION** • Always reconnect all disconnected connectors. • Clear the DTC from the PCM memory using the M-MDS. (See CLEARING DTC [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) • Perform the KOER self test. (See KOEO/KOER SELF TEST [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) • Is the same DTC present?	Yes	Replace the PCM, then go to the next step. (See PCM REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
		No	Go to the next step.
3	**PURPOSE: VERIFY IF THERE IS ANY OTHER MALFUNCTION** • Is any other DTC or pending code stored?	Yes	Go to the applicable DTC inspection. (See DTC TABLE [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
		No	DTC troubleshooting completed.